# Forward-Registered, Auditable LLM-Assisted Research: A Reliability Methodology, with a Capital-Markets Testbed

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**Abstract**

Large language models are increasingly used as decision engines for open-ended research, but they are unreliable in characteristic ways: they generate plausible but unsupported claims, amplify narrative priors, over-admit candidates, and — when evaluated retrospectively — overfit data that postdates their training. We study this as a problem of **AI reliability and auditability**, using open-ended thematic investment research as a concrete, adversarial testbed. We present **a forward-registered, agent-assisted research framework** that wraps LLM-driven discovery in deterministic gates and a forward pre-registration protocol, so that a hypothesis can be admitted to a forward-evaluation portfolio only after surviving falsifiable, dated-evidence checks and being evaluated *forward* on paper trading rather than by a fragile backtest.

The framework contributes four components.

1. A **framework**: a domain-agnostic bottleneck graph in which nodes are hard-to-replicate chokepoints, physical or otherwise, and adding a domain is mostly a matter of configuration (a new bottleneck map) rather than new modeling code, with a decomposed confidence model (bottleneck $\times$ purity $\times$ demand $\times$ valuation $\times$ crowding).

2. A **protocol** for disciplined judgment: deterministic gates, source-tier rules, a bear case written before any recommendation, overlap penalties, and government policy entered as *dated evidence and a reversal trigger, never as a multiplicative factor* — together an explicit mechanism for *rejecting* candidates, not merely surfacing them.

3. **Forward-QPOP**, a **hash-chained**, append-only pre-registration ledger for hypothesis admission, belief updates, and exits — the contribution is the *forward-locked hypothesis contract* (dated evidence + entry/update/exit triggers committed before the evaluation window), not the tamper-evidence alone.

4. A **pre-registered evaluation with pilot evidence** — a low admission rate, a held-out rejection-quality audit, discipline ablations against ungated baselines, and a cross-domain portability probe — released as an open framework, with forward outcomes left to the evaluation window.

*The author maintains a private research book applying this method and may hold positions in names within the example domains discussed here. Example names are illustrative of the decision taxonomy — not disclosures, endorsements, or recommendations. See §"Ethics and Responsible Use."

Rather than a return claim, our central contribution is methodological: **this validates process discipline, not investment profitability** (which awaits the forward window). We test whether a disciplined, forward-registered agent **rejects most of the candidates it surfaces** *for defensible reasons*. We report *pilot* discipline metrics from early candidate batches — a low admission rate and a **held-out LLM-auditor rejection-quality check**. In exploratory system contrasts on a 38-candidate batch, an ungated LLM screener — free to reject — admitted **38/38** while the full pipeline admitted **0/38**; over the frozen forward window the run ledger records **0 admissions across 9 discovery rounds and 110 gate-pass evaluations** — "no action" is the modal outcome. On a documented, seeded sample of $n$=40 rejections the held-out bull-only auditor **agreed with 31/40 (0.775)** of the engine's rejections, which **does not meet the** $\geq 0.80$ **criterion later registered for the prospective H5 window**; we report that gate failure as a result. The earlier pilot's 0.93 (an undocumented $n$=14) was a small-sample optimistic read: the documented estimate lies inside the pilot's own interval, so nothing is contradicted, but the pilot headline did not survive a 2.86× larger, documented sample (the auditor-flag count rose 9×, from one to nine). A full-record LLM adjudicator overrode all nine bull-only flags (9/9); because auditor and adjudicator are related models judging under deliberately asymmetric information, we report this as **model disagreement, not rejection ground truth** — rejection quality remains unresolved pending a pre-specified human audit. These are LLM-on-LLM diagnostics that *motivate* the pre-registered forward evaluation, not confirm it; forward returns are not yet observed. The method makes each admitted position explainable from its content-hashed ledger entry, whereas ungated LLM screeners lack an explicit rejection mechanism. We release the framework, a worked AI-supply-chain example with a real candidate flow, and a template for new domains.[1]

**AI disclosure.** Large language models are used in this work in two roles: as the *object of study* — the agent-assisted research workflow this paper evaluates — and as *tools* in preparing the manuscript (drafting, code, and editorial assistance). The author reviewed and is solely responsible for all claims, citations, code, and conclusions.

## 1 Introduction

Large language models have made one kind of work almost free: open-ended, agent-assisted research. A practitioner can surface dozens of candidate hypotheses in minutes — in any domain, and, in the testbed of this paper, thematic-investment ideas such as bottleneck plays, supply-chain theses, or policy-driven opportunities. The danger, however, is symmetric: the same fluency that makes LLMs productive *sourcing* tools makes them unreliable *decision engines*. LLMs are rewarded, by training and by user expectation, for producing confident, coherent narratives. Left ungated, an LLM screener over-admits by construction—it is optimised to find ideas, not to refuse them—and the narratives it generates are difficult to audit, reproduce, or falsify after the fact.

Existing work on financial LLM agents has concentrated on improving recommendation quality: better prompts, chain-of-thought reasoning, retrieval-augmented generation, or multi-agent debate (Xiao et al., 2024; Du et al., 2024). These advances are real, but they address the wrong bottleneck. The limiting constraint for a practitioner deploying LLM-assisted research is not the *quality of individual suggestions*; it is the absence of a disciplined, auditable mechanism for *rejecting*

[1] Code, JSON schemas, synthetic fixtures, and this paper's source: `https://github.com/yixingz3/qpop`. The ledger, hash chain, timestamp anchor, e-value module, prompts, and example domains are public; the live gate/scoring/domain-templating engine is private — see § Reproducibility. The `forward-qpop` ledger is also on PyPI (`pip install forward-qpop`).

the majority of what the model surfaces. Without such a mechanism, every narrative becomes a plausible trade, and the researcher is left arbitrating between fluent confabulations rather than falsifiable hypotheses. What is needed is not better prompts—it is **auditable restraint**.

This paper describes a methods framework that enforces that restraint through five interlocking mechanisms: a domain-portable **bottleneck graph** that decomposes a thematic thesis into hard-to-replicate chokepoints; a **confidence decomposition** that derives a position tier from scored node-and-ticker attributes rather than assertion:

$$\text{confidence} = \underbrace{\text{bottleneck\_score}}_{\text{node: how binding}} \times \underbrace{\text{exposure\_purity}}_{\text{ticker: proxy quality}} \times \underbrace{\text{demand\_score}}_{\text{node: order visibility}} \times \underbrace{\text{valuation\_adjustment}}_{\leq 1} \times \underbrace{\text{crowding\_adjustment}}_{\leq 1}$$

a **cheap-source → deterministic-gate → expensive-evaluate** funnel that concentrates costly inference on the small fraction of candidates that survive reproducible, LLM-free gate checks; a **bear-case-before-recommendation** evaluation discipline that counteracts sycophancy; and a **forward-QPOP pre-registration ledger** (QPOP is a coined protocol name, not an acronym) that commits hypotheses, dated evidence, and measurable exit triggers *before* any capital-allocation claim, foreclosing HARKing and backtest overfitting.

The pilot empirical result reported in this draft is deliberately not a return claim. It is the **admission rate**: a disciplined agent *rejects most of what it surfaces*. In our initial pilot batch, fewer than 10% of *evaluated finalists* (and a lower share of all LLM-surfaced candidates) were admitted to the forward ledger; the remainder were rejected via deterministic gates or bear-case evaluation. On a later documented, seeded sample of 40 rejections the held-out bull-only *LLM auditor* agreed with 31/40 (0.775) of the engine's rejections, which does *not* clear the $\geq 0.80$ criterion later registered for the prospective H5 window — a result we report rather than bury; a full-record LLM adjudicator overrode every one of the nine auditor flags (model disagreement under asymmetric information — an LLM-on-LLM diagnostic, not a human audit or ground truth). We report this restraint figure alongside exploratory system contrasts—ungated screener, debate-only, no-QPOP-lock, no-overlap-penalty—in which every weakened-discipline configuration admits far more than the full pipeline (a configuration-level observation; no single component's causal contribution is identified). This is a *methods and systems* paper: the contribution is a workflow whose released scaffold — the ledger, hash chain, anchor, e-value module, reference prompt templates (sanitized; the H2 arm prompts released verbatim), schemas, and domain template — any practitioner can adopt, audit, and extend. Supplying a new `bottleneck_map.yml` reuses the public discipline, ledger, and template for one's *own* funnel, implemented from the released prompts and reference gate specification; it does not run or reproduce *our* production funnel, whose gate/scoring/domain-templating engine is private (§ Reproducibility).

# 2 Related Work

## 2.1 LLM and Agentic Systems for Investing and Financial Research

Domain-adapted language models have become the baseline for financial NLP. Wu et al. (2023) introduced BloombergGPT, establishing strong financial-NLP performance but offering no decision discipline or rejection mechanism (Wu et al., 2023). Yang et al. (2023) released FinGPT as an open-source counterpart (Yang et al., 2023). Moving to autonomous agents, Ding et al. (2024) surveyed LLM trading agents, finding that nearly all evaluation is in-sample relative to the model's training cutoff (Ding et al., 2024). Xiao et al. (2024) built TradingAgents, a bull/bear-debate multi-agent framework evaluated on backtest Sharpe (Xiao et al., 2024); Zhang et al. (2024) contributed FinAgent, a multimodal agent reporting large backtested profit improvements (Zhang et al., 2024). None impose a pre-registered hypothesis contract, an explicit admission-rate target, or forward validation; their headline metrics are backtest Sharpe ratios rather than the disciplined rejection of poor ideas. Two 2025–2026 results sharpen exactly this point. Ye et al. (2026), in *The Alpha Illusion*, argue that reported alpha from end-to-end LLM trading agents should not be treated as deployment evidence until it survives structural-validity tests (temporal integrity, frictions, robustness, calibration, execution), and recommend positioning the LLM as an upstream information interface feeding independent risk/execution modules — not as the decision engine (Ye et al., 2026). Lee et al. (2025), *Your AI, Not Your View*, give the first quantitative analysis of LLMs' inherent investment biases (large-cap and contrarian preference, and confirmation bias that makes models cling to an initial judgment against counter-evidence) (Lee et al., 2025). Both motivate our central design choice: the LLM *sources* hypotheses, but deterministic gates and a forward contract, not the model's own confidence, decide whether anything is admitted.

## 2.2 LLM Failure Modes Relevant to Research

Three failure families matter here. **Hallucination/confabulation:** models fabricate plausible facts beyond their training distribution, carrying direct fiduciary risk. **Sycophancy:** Malmqvist (2024) catalogued how LLMs systematically agree with user-expressed views, generating false confidence — critical when the model is asked to rate a thesis it also generated (Malmqvist, 2024). **Lookahead leakage:** Gao et al. (2025) showed that a positive lookahead-accuracy correlation is direct evidence the model saw the answer in pre-training (Gao et al., 2025); Li et al. (2025b) showed strong backtested LLM-agent returns collapse out-of-sample once leakage is controlled (Li et al., 2025b). Benhenda (2026) contributes *Look-Ahead-Bench*, a regime-generalization benchmark separating genuine predictive skill from memorized patterns, finding standard LLMs exhibit significant look-ahead bias while point-in-time models do not (Benhenda, 2026); Roy and Roy (2026), *MemGuard-Alpha*, show in-sample accuracy rises with memorization contamination while out-of-sample accuracy falls, and use cross-model disagreement as a contamination signal (Roy and Roy, 2026). Surveying the field, Kong et al. (2026) review 164 studies (2023–25) and find no single bias — lookahead, survivorship, narrative, objective, or cost — is explicitly addressed in more than $\sim$28% of them, and propose a (retrospective, reviewer-facing) structural-validity checklist (Kong et al., 2026). At the factor level, Harvey et al. (2016) showed most claimed return factors are likely false discoveries from multiple comparisons, requiring a $t > 3.0$ threshold (Harvey et al., 2016). Our framework treats these as first-class *design* constraints rather than post-hoc review items: deterministic gates before any expensive-model judgment, bear-case-before-recommendation, and forward (not retrospective) validation.

## 2.3 Pre-Registration and Research-Integrity Methods

Nosek et al. (2018) argued the core value of preregistration is the sharp separation of prediction from postdiction (Nosek et al., 2018). Casey et al. (2012) gave the canonical economics pre-analysis-plan application (Casey et al., 2012). In finance, McLean and Pontiff (2016) documented that published return predictors decay $\sim$26% out-of-sample and $\sim$58% post-publication, much of it data-mining bias (McLean and Pontiff, 2016); work on pre-analysis plans finds they suppress p-hacking only with a *detailed* plan, not registration alone (Brodeur et al., 2024). Closest to our setting, Hofman et al. (2023) adapt clinical-trial pre-registration to predictive modeling, naming three reproducibility failures (overlooked contextual factors, data-dependent decisions, unintentional test-set reuse) and giving a lightweight ML pre-registration template (Hofman et al., 2023). Our **forward-QPOP** protocol extends this tradition with two elements that template lacks: (i) a domain-specific *bottleneck* schema that makes pre-registration portable across investment themes, and (ii) a *forward* outcome commitment — every hypothesis is content-hashed before the market opens, evaluated on live paper-trading fills rather than historical simulation, and the ledger is append-only — closing the loophole of testing on periods that postdate the model's training cutoff.

## 2.4 Multi-Agent LLM Orchestration and Verification

Du et al. (2024) formalized multiagent debate to improve factuality (Du et al., 2024). Li et al. (2025a) surveyed "LLM-as-a-judge," identifying self-enhancement bias when generator and judge are the same model (Li et al., 2025a). Crucially, more judges do not mean less bias: Ma et al. (2025) show debate-based multi-agent judging *amplifies* position, verbosity, chain-of-thought, and bandwagon biases after the first round (Ma et al., 2025), and Zhao et al. (2026) measure sycophancy in *agentic financial* pipelines, finding most models abandon a correct analytical position once a user preference is expressed (Zhao et al., 2026). These results argue that an adversarial frame must be *structurally enforced*, not left to emergent debate — which is exactly our bear-case-before-recommendation rule. The most methodologically adjacent work is POPPER (Huang et al., 2025), which uses LLM agents to run sequential *falsification* experiments with Type-I error control (Huang et al., 2025). We differ on three axes a reviewer will probe: (i) **forward lock vs. retroactive test** — POPPER validates a hypothesis against *already-existing* data and cannot stop a researcher from running it after seeing the outcome, whereas our hypotheses are content-hashed and dated *before* the out-of-sample window opens; (ii) **discovery vs. validation** — POPPER receives a formed hypothesis, while our workflow begins before the hypothesis exists, sourcing candidate themes through the bottleneck schema; (iii) **domain portability** — our schema carries finance-specific, reusable evidence fields (chokepoint dimensions, dated triggers, source tiers) rather than generic statistical falsification. Our design also keeps the cost discipline: a *cheaper* model gates and a *more expensive* model evaluates only gate-passers, not all candidates.

## 2.5 Thematic / Supply-Chain / Chokepoint Investing; Factor Crowding

Factor crowding is well-established: Volpati et al. (2020) quantified crowding in standard factors (Volpati et al., 2020); Lou and Polk (2022) showed "comomentum" predicts the reversal of crowded momentum (Lou and Polk, 2022); Harvey et al. (2016) frame the broader factor-zoo / false-discovery problem (Harvey et al., 2016). Our bottleneck-graph model formalizes chokepoint identification as a structured schema (supply concentration, demand inelasticity, substitutability, pricing power), and the confidence decomposition tracks crowding and valuation as *gating* dimensions, so a thesis can be rejected on crowding grounds before any simulation.

## 2.6 Trustworthy / Responsible AI in Finance

Fritz-Morgenthal et al. (2022) connected trustworthy-AI principles (fairness, explainability, robustness, accountability) to financial risk management, arguing self-learning models must archive full data and metadata after material changes (Fritz-Morgenthal et al., 2022). Khatchadourian (2026) shows empirically that decision determinism and task accuracy are *uncorrelated* in tool-using LLM financial agents and that schema-first architectures give the most audit-compliant consistency (Khatchadourian, 2026), while Buscemi et al. (2026) map EU AI Act high-risk requirements to concrete technical-verification activities and document the gap between regulatory intent and current practice (Buscemi et al., 2026). Regulatory direction (EU AI Act 2024; NIST AI RMF) treats high-risk AI as requiring documented, auditable decision trails — a standard backtest-centric agents do not meet. Our content-hashed, timestamped, forward-validated, human-readable ledger makes the system's reasoning inspectable independently of the LLM that produced it.

### 2.7 Live LLM-Trading Benchmarks and Auditable Trade Logs

A reviewer will rightly note that neither *forward evaluation* nor *tamper-evident logging* is, on its own, novel — so we state precisely where the line is. Qian et al. (2025), *When Agents Trade*, introduce the Agent Market Arena, a **lifelong, real-time multi-market benchmark** for LLM trading agents, finding that agent architecture drives risk behavior more than the base LLM (Qian et al., 2025). Chen et al. (2025), *StockBench*, give a contamination-free sequential trading benchmark and find most LLM agents fail to beat buy-and-hold (Chen et al., 2025). On the engineering side, the open-source `llm-quant` system records LLM paper-trades in a **SHA-256 hash-chained, git-tracked ledger** (45ck, 2025). We borrow from this last design — our Forward-QPOP ledger is likewise hash-chained — but our contribution is *not* the immutable log. The distinction, which we make central: these systems evaluate or record **what an agent *did*** in the market; ours forward-**locks a researcher-level hypothesis** (the binding-chokepoint claim, dated evidence, and entry/update/exit triggers) to a hash-stamped record **before the evaluation window opens**. Live evaluation tells you the outcome; the forward lock makes the *prediction* auditable against that outcome, closing the gap that lets backtest- or benchmark-based claims be rationalized after the fact.

### Gap We Fill

Existing work addresses individual validity threats in isolation: lookahead-bias diagnostics and benchmarks (Gao et al., 2025; Benhenda, 2026), structural-validity checklists applied *post-hoc* by reviewers (Kong et al., 2026), automated falsification for *already-formed* hypotheses (Huang et al., 2025), output-layer auditability for tool-using agents (Khatchadourian, 2026), and live trading benchmarks or hash-chained trade logs that record *executions* rather than *pre-registered predictions* (Qian et al., 2025; Chen et al., 2025; 45ck, 2025). To our knowledge (a literature search through June 2026, across the LLM-agent, financial-ML, and AI-auditability venues cited above), no prior system ties these into a *prospective, forward-locked* workflow that unifies all four capabilities: (a) agentic open-ended hypothesis discovery across thematic domains; (b) deterministic gating that enforces source-tier, purity, and overlap rules *before* any expensive model is invoked; (c) **forward pre-registration — not backtesting — as the evidentiary standard**, immutable once content-hashed and dated; and (d) a portable cross-domain bottleneck schema that makes the process reproducible and auditable without relying on any single LLM instance or training snapshot. The combination is what converts the LLM from a decision engine into an auditable sourcing tool. The closest peer, POPPER (Huang et al., 2025), shares the falsification spirit but validates a static hypothesis against existing data; it has neither the forward temporal lock (which prevents backfitting) nor the discovery phase nor the finance-portable schema; Table 1 summarizes the distinction.

## 3 Problem Formulation

We state the target abstractly before specializing it to the capital-markets testbed. A **candidate** $c$ is a proposed research claim paired with a possible *realization vehicle*. An LLM-assisted *source*

| System | Starts before hypothesis? | Forward-locks the claim? | Primary reliability target |
|---|---|---|---|
| POPPER (Huang et al., 2025) | No | No | Type-I error validating a formed hypothesis |
| Live-trading benchmark (Qian et al., 2025; Chen et al., 2025) | Usually | No | agent trading outcome |
| Hash-chained trade log (45ck, 2025) | No | Logs actions | tamper-evidence of executions |
| **Forward-QPOP (ours)** | **Yes** | **Yes** | over-admission & auditable restraint |

Table 1: Where Forward-QPOP sits among adjacent systems: to our knowledge, the only one that begins *before* a hypothesis is formed and *forward-locks* the claim before its outcome (search scope: § Gap We Fill).

process surfaces candidates in volume; the framework maps each to one of three decisions,

$$d(c) \in \{\textsc{reject},\ \textsc{watchlist},\ \textsc{admit}\}.$$

The realization vehicle is domain-specific (Table 2): in this paper's capital-markets testbed it is a listed security; in an ML-evaluation setting it may be a model, dataset, benchmark, or experimental configuration (a worked non-finance example ships in the released repository).

| General concept | Capital-markets testbed | ML-evaluation example |
|---|---|---|
| candidate | (chokepoint node, ticker) claim | (method, benchmark) claim |
| realization vehicle | listed security | model / dataset / configuration |
| deterministic gate | tradeability, liquidity, overlap | eligibility, de-duplication |
| bear-case evaluation | over-payment / crowding check | ablation / threats-to-validity |
| admit | open a position | pre-register the eval claim |
| exit / update trigger | price or lead-time signal | metric threshold |

Table 2: The pipeline is domain-agnostic; the worked example specializes the generic "candidate" to a (node, ticker) pair.

The primary reliability failure of an ungated LLM workflow is **over-admission**: admitting plausible but weak candidates that a disciplined process would reject. We therefore measure not only *how few* candidates a pipeline admits, but whether its rejections are *defensible* and whether admission *rises* under weakened discipline configurations. Three metrics make this precise.

**Definition 1** (Admission rate)**.** *The fraction of evaluated finalists the pipeline admits, admitted/finalists. A low admission rate is necessary but not sufficient for reliability — a process can reject everything by being arbitrarily conservative.*

**Definition 2** (Rejection precision)**.** *On a held-out audit of rejected candidates, the fraction judged* justified*, justified/audited. This guards against arbitrary conservatism.*

**Definition 3** (Over-admission rate (OAR))**.** *For a fixed candidate set, the share that an ablated or ungated process* admits *but the full discipline* rejects — *a reference-relative admission-disagreement quantity (formalized, with its mirror direction, as rOAR/rOMR in § Results). It is the portable reliability target this framework minimizes; attributing it to any* single *removed discipline requires matched, nested arms, which the pilot does not have.*

A reliable workflow drives OAR down *while* keeping rejection precision high — defensible restraint, not a rejection machine. Algorithm 1 is the decision procedure; the remainder of the paper instantiates and evaluates it in the capital-markets testbed, while the ledger and disciplines are domain-agnostic by construction.

**Algorithm 1: The admission ladder.** Input: candidate $c$. Output: $d(c)$. Apply in order; stop at the first failure → REJECT (the modal outcome).

1. **Story or claim?** State $c$ as something falsifiable, with a mechanism — or reject.
2. **Deterministic gates** (no LLM): eligibility, and *replace-don't-stack* on duplicates.
3. **Bear case first**: the strongest case *against*, written before any recommendation.
4. **Source tiers**: primary > secondary > market-implied > tertiary; tertiary alone never moves a conclusion.
5. **Forward pre-registration**: claim + dated evidence + exit triggers, hash-chained *before* the window.
6. **Forward, not backtest**: a backward test of an LLM-scored process is structurally invalid.
7. **On balance**: after overlap, cost, and uncertainty, does admitting beat doing nothing? Otherwise WATCHLIST.

# 4 Framework

We now instantiate the general pipeline of §3 in the capital-markets testbed. The framework turns open-ended, LLM-driven discovery into *disciplined, auditable, falsifiable* research by composing five mechanisms. Each exists to neutralize a specific, well-documented failure mode of using LLMs for investment research.

| **Mechanism** | **Failure mode it neutralizes** |
|---|---|
| Bottleneck graph + confidence decomposition | "right theme → buy the ticker"; theme-chasing |
| Cheap-source → deterministic-gate → expensive-evaluate funnel | cost blow-up; shallow, ungrounded judgment |
| Bear-case-before-recommendation | sycophancy; one-sided narratives |
| Forward-QPOP pre-registration | HARKing; data-snooping; backtest overfitting |
| No-stories discipline + admission restraint | confabulation; over-admission; narrative drift |

## 4.1 The bottleneck graph (the portable thesis)

A domain is a **graph of hard-to-replicate chokepoints, physical or otherwise**. Nodes are bottlenecks (a step that is hard to bypass, capacity-constrained, supplier-concentrated); edges encode dependency. The graph is domain-agnostic: AI accelerators today, rare-earth refining / power transformers / LNG / uranium / copper tomorrow — each is *a different* `bottleneck_map.yml`, not a different codebase.

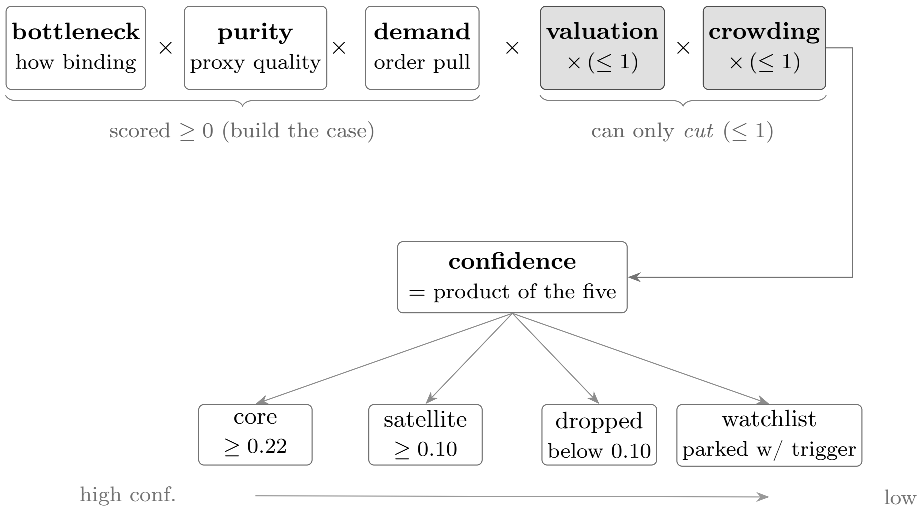


Figure 1: The confidence decomposition. Three evidence-scored factors and two multiplicative penalties capped at 1 (a crowded or richly-valued name can only lose confidence, never gain it) combine into a single score, from which the position tier — and thus sizing (core $\geq 0.22$, satellite $\geq 0.10$) — is *derived*, not asserted.

Each node carries: a physical rationale, `bottleneck_dims` (`physical_indispensability`, `substitutability`$^{-1}$, `capacity_lead_time`, `supplier_concentration`, `pricing_power`), a `demand_score`, node-level `valuation_adjustment` / `crowding_adjustment`, and **measurable `exit_triggers`, each with a checkable** `data_source` (an admission rejects any trigger you cannot actually check).

**Scoring unpriced scarcity: the valuation and crowding adjustments**

A binding chokepoint is necessary but **not sufficient**: what the engine must score is how much of the scarcity is *not yet priced in.* This is what the `valuation_adjustment` and `crowding_adjustment` factors capture — they discount a constraint the market has already learned, so the engine prefers chokepoints whose binding is still being discovered over ones whose schedule everyone already trades. (The underlying "uncertainty about the constraint" principle is a conceptual illustration — developed, with examples, in the Discussion — not a contribution of this methods framework.)

## 4.2 Confidence decomposition (kills theme-chasing)

$$\text{confidence} = \underbrace{\text{bottleneck\_score}}_{\text{node: how binding}} \times \underbrace{\text{exposure\_purity}}_{\text{ticker: captures economics}} \times \underbrace{\text{demand\_score}}_{\text{node: order visibility}}$$
$$\times \underbrace{\text{valuation\_adjustment}}_{\leq 1\text{: penalize priced-in}} \times \underbrace{\text{crowding\_adjustment}}_{\leq 1\text{: penalize crowded}}$$

A correct node with a poor proxy (low purity) or a crowded entry yields *low* confidence. Tier is **derived** from confidence (e.g. core $\geq 0.22$, satellite $\geq 0.10$, below $\rightarrow$ dropped), not asserted (Figure 1).

### Policy is *evidence*, not a factor

Government policy (subsidies, awards, export controls, strategic-autonomy programs) is a real, sometimes durable driver — but it enters as **dated evidence + a monitor trigger**, never as a multiplicative `policy_adjustment`. A multiplier manufactures a number from a *forecast* (is it durable? bipartisan?) — the no-stories failure mode in a lab coat. Two cases instead:

- **Tailwind on a name with a real chokepoint** → record a (loader-inert) `policy:` evidence block + a wired `policy_reversal` exit trigger (clawback / milestone miss / reversal). It does **not** resize the name.
- **Underdog whose thesis *is* policy** → a *separate, capped* sleeve, admitted via the normal gates, never core.

**Fact vs announcement:** a *signed, dated, material* award is citable evidence; a political headline / pledge is watchlist-context only. The clinching lesson: a multiplier *rewards the existence* of policy — but a signed award that has *already fired* (in the price) with its clawback removed is sunk information turned into a *liability*; the evidence-and-trigger framing forces the right questions (*has it already fired? what would reverse it? does it unlock capacity?*).

## 5 Forward-QPOP Protocol

**QPOP** is a forward hypothesis ledger. It adapts pre-registration and registered-reports discipline from empirical science into agent-assisted investment research, *without* relying on a backtest.

### Why forward, not backtest

For a young or structurally-shifting universe (recent IPOs, a thesis the market is actively re-rating), a long backtest is fragile: regime change, survivorship, and lookahead leakage make it easy to overfit a story. Pre-registration replaces "fit the past" with "commit, then observe the future." The ledger is the contract against **HARKing** (hypothesizing after results are known) and **data-snooping**.

For an *LLM-scored* engine the objection is sharper than fragility: a backward test is **structurally invalid**, not merely noisy — and three properties defeat it, each sufficient on its own. **(i) No information is injected backward.** The engine is a deterministic composition of LLM judgments, so retrodicting historical names measures the model's *hindsight*, not the engine's edge. **(ii) The labels are already in the model.** The outcomes one would test against have settled and are present in the model's training corpus, so there is no clean train/test split: the label leaks into the feature through pretraining. **(iii) The features *are* the fingerprint.** The very attributes the engine scores — which chokepoint, what purity, whose capacity — uniquely identify the company and hence its known outcome, so anonymizing the inputs cannot restore a blind test. The consequence is not "prefer forward"; it is that *no* backward procedure can evidence the engine's predictive signal — a backward run is at best an internal-consistency check. This is the engine-level analogue of the HARKing argument the ledger already enforces at the hypothesis level, and it is why we operationalize forward evaluation with the registry below rather than a historical analog study.

### The forward-scoring registry (engine-level evaluation)

The hypothesis ledger validates individual *admissions* forward. To evaluate the *scoring engine itself* — whether a higher `final_confidence` ranks a higher forward return — we keep a separate

**append-only forward-scoring registry**. At a cohort date we freeze every scored name's effective tier (core/satellite, assigned by the *production* tiering rule on post-refresh values, not a bare threshold) together with its decomposed confidence, and then let outcomes settle *after* the model's knowledge. Because the signatures are recorded forward and the outcomes are unanticipated, the test is honest in exactly the way a backward retrodiction (§ above) is not. The registry is **provisional** until it accrues power — on the order of dozens of name–horizon observations spanning both high- and low-confidence names — so it is a multi-year instrument, reported with the structural-validity checklist, never a near-term gate. A subtle requirement: the cohort must be labeled with the same tiering function the live book uses (including any hold/override path), or the registry's own labels disagree with the book it claims to validate.

## Entry types

1. **Admission** — a node or ticker enters the map (especially at meaningful weight) *only* via a pre-registered hypothesis: the binding-chokepoint claim, dated evidence and source tier, the expected return driver, the decomposed confidence, and **measurable exit triggers committed in advance**. No pre-registration → no capital.

2. **Belief update** — every confidence change is a recorded, **content-hash-protected** posterior update with cited evidence. *Tertiary sources alone cannot move confidence.*

3. **Outcome** — when a trigger fires or the thesis ages into consensus, the entry is closed **Supported / Weakened / Falsified**, with the observed metrics. **Closed entries are immutable — open a new id to revise.**

### Entry schema (illustrative)

```yaml
id: H-<DOMAIN>-<NN>
created: <date>
status: open            # open | supported | weakened | falsified
role_admitted: satellite   # never straight to core
claim: "<the binding-chokepoint claim, in one sentence>"
mechanism: "<why it is hard to bypass>"
decomposed_confidence:
  bottleneck_score: 0.0
  exposure_purity: 0.0
  demand_score: 0.0
  valuation_adjustment: 0.0
  crowding_adjustment: 0.0
  final_confidence: 0.0
objective_gates:
  { tradable: true, min_price: true,
    min_dollar_volume: true, overlap_penalty: 0.0 }
exit_triggers:
  - { id: supply_easing,
      metric: "...", op: "<",
      data_source: { name: "...", tier: secondary } }
  - { id: demand_slowing,
      metric: "...", op: cut,
      data_source: { name: "...", tier: primary } }
  - { id: crowding,
      metric: "price vs estimate-revision gap", op: ">",
      data_source: { tier: market_implied } }
prior_confidence: 0.0
evidence:
  - { summary: "...", tier: primary, date: <date>, url: "..." }
content_hash: "<hash of the frozen fields>"
prev_hash:    "<content_hash of the previous ledger entry>"
entry_hash:   "<sha256(content_hash + prev_hash)>"
```

### The hash chain

The load-bearing fields (`claim`, `exit_triggers`, `decomposed_confidence`, `prior`) are hashed at creation (`content_hash`). Each entry **also binds the previous entry's hash** (`prev_hash`), so the ledger is a **hash chain**:

$$\texttt{entry\_hash} = \text{sha256}(\texttt{content\_hash} \,\|\, \texttt{prev\_hash})$$

not merely a set of independently-hashed rows. Two properties follow:

- **Per-entry integrity:** any later edit to a *closed* entry breaks its `content_hash` and is rejected — when externally timestamp-anchored, an external reviewer can verify the hypothesis was registered *before* the outcome.
- **Append-order integrity:** because each entry commits to its predecessor, you cannot silently insert, delete, or re-order a past entry without breaking every `entry_hash` after it. This is the difference from a bare "immutable trade log" (e.g. a hash-chained *execution* ledger): here the chained object is the *pre-registered hypothesis contract* — claim, dated evidence, and entry/update/exit triggers committed before the evaluation window — so the chain proves *what was predicted and its append order* (and, with the package's optional OpenTimestamps anchor, the before-outcome timing), not just what was traded.

*Design note:* the hash-chain construction is borrowed from tamper-evident LLM trade logs; the contribution here is chaining the *hypothesis* (with its forward triggers), not the trade.

### Dated admission, update, and exit triggers

Each trigger committed at admission has three mandatory fields:

| Field | Type | Purpose |
|---|---|---|
| `id` | string | unique label within the entry |
| `metric` | string | the observable quantity checked against the threshold |
| `op` | string | comparison operator or named rule (e.g. `<`, `cut`) |
| `data_source.name` | string | named source for the metric |
| `data_source.tier` | enum | `primary` \| `secondary` \| `market_implied` |

*Admission trigger (entry):* hypothesis registered before any capital deployed.
*Update trigger (belief update):* posterior revision with cited evidence; tertiary sources alone blocked.
*Exit trigger:* pre-committed rule; outcome recorded as Supported/Weakened/Falsified.

### Forward validation and posterior recording

After each forward observation window:

- Did a pre-registered trigger fire? → reduce/exit per the rule, record the outcome. *Never a story.*
- Did the thesis age into consensus (valuation/crowding adjustments collapse)? → **Weakened**.
- Did the chokepoint hold and the driver play out? → **Supported**.

The accumulated (hypothesis → decision → forward outcome) tuples form a reusable, auditable **dataset** — both the empirical core of the methods paper and the fuel for calibrating the gates on realized outcomes (the self-improvement loop).

## 6 System Implementation

The framework's five mechanisms are realized as a staged, model-tiered *cascade* — cheap, deterministic filters first, the expensive model last.

### The Source→Gate→Triage→Evaluate→Adjudicate Cascade

The discovery pipeline is a five-stage funnel in which each tier hands the next a smaller, higher-value candidate set:

| `SOURCE` | → | `GATE` | → | `TRIAGE` | → | `EVALUATE` | → | `ADJUDICATE` |
|---|---|---|---|---|---|---|---|---|
| cheap LLM | | no LLM | | cheapest LLM | | mid LLM | | expensive LLM |
| wide, deduped | | deterministic | | drop low-purity | | bear-case-first | | admit-flags only |

- **SOURCE** (cheap model): high-volume, low-judgment candidate cards with dated, source-tiered evidence — one lens per node, plus gap / social / policy / **literature** lenses. **Deduped against a persistent "seen" set** (every previously-held / watchlisted / rejected name): the lenses skip a decided name unless it shows a *material, dated* change, so the expensive sourcing tokens are spent on genuinely new candidates, not re-researching the book.
- **GATE** (no LLM): tradeability + liquidity + **overlap penalty** vs. the existing book. Pure, reproducible, cheap. A name failing the gate is watchlist-only, never "free capacity."
- **TRIAGE** (cheapest LLM): a quick purity pre-filter that drops the obvious non-starters (diversified-giant-tiny-segment / commodity price-taker / pre-revenue hype) *before* the expensive bear-case pass. **Safe by asymmetry** — a wrong drop only watchlists a name (a delay), and a wrong pursue costs one cheap evaluation that watchlists it anyway.
- **EVALUATE** (mid model, *triage survivors only*): deep fit + a **bear case written before the recommendation** + a falsifiable decision.
- **ADJUDICATE** (expensive model, *admit-flags only*): a capital-at-risk re-judgment of the names the EVALUATE pass flags to admit/replace. Reserve the scarce model for exactly the irreversible call.

Model allocation is a tunable cascade: each tier hands the next a smaller, higher-value set, so the cheapest models do the volume and the expensive model only ever judges admits. The asymmetry is safe throughout — a false-watchlist merely delays a name, while a false-admit is caught downstream.

## Confidence Decomposition

A correct bottleneck node with a poor proxy ticker (low exposure purity) or a crowded entry yields *low* confidence. The tier assigned to a name is **derived** from the confidence score (e.g. core $\geq 0.22$, satellite $\geq 0.10$, below $\rightarrow$ dropped), not asserted by the analyst. The decomposition is:

$$\text{confidence} = \underbrace{\text{bottleneck\_score}}_{\text{node: how binding}} \times \underbrace{\text{exposure\_purity}}_{\text{ticker: captures economics}} \times \underbrace{\text{demand\_score}}_{\text{node: order visibility}} \times \underbrace{\text{valuation\_adjustment}}_{\leq 1\text{: penalize priced-in}} \times \underbrace{\text{crowding\_adjustment}}_{\leq 1\text{: penalize crowded}}$$

## No-Stories Discipline and Forward Validation

- **Source tiers:** primary (filings/transcripts) > secondary (reputable trade press) > market-implied (price/volume) > tertiary (social — *idea seed only*, never establishes confidence). A confidence move requires $\geq$ secondary or market-implied corroboration.
- **Overlap penalty:** every candidate is judged *against the book you already hold* — a great theme that duplicates a held bet must **replace, not stack** (gate = comparison, not veto).
- **Structural vs. cyclical:** distinguish a hard chokepoint (few suppliers, multi-year capex) from a commodity-cycle peak riding a bullwhip; the latter is the trap that *looks* like the winner.
- **A position changes only on** a fired pre-registered trigger, a binding cap, or a risk limit — **never a narrative.**

- **Validation is forward:** track-record metrics are judged on *conservative shadow fills* (opening print + assumed slippage), benchmark-relative, with the accumulated ledger as the dataset. The system is optimized for *auditability*, not cleverness — every weight must be explainable by score $\times$ purity $\times$ adjustments $\times$ caps $\times$ drift, or no position is taken.

### Why the Admission Rate Is the Headline — and Why It Is Not Enough Alone

A naïve LLM screener over-admits by construction — it is rewarded for *finding* ideas. This framework is rewarded for *refusing* them: most candidates are on-theme yet fail "does this improve the book after overlap, purity, valuation, and crowding?" The low admission rate, achieved by pre-committed gates rather than after-the-fact judgment, is one measurable contribution.

But a low admission rate is **not, by itself, evidence of quality** — a system can reject 95% of ideas by being arbitrarily conservative. So restraint is reported with two companions:

- **Rejection quality.** A held-out auditor (an LLM and/or a human), shown only the *bull* case and the decision — **not** the engine's bear case — labels each sampled rejection JUSTIFIED or a FALSE REJECTION, with a category (low-purity / duplicate-overlap / valuation-or-crowding / commodity-cycle / pre-revenue-or-hype / unverifiable-or-untradeable). The headline pairs *admission rate* (how much is rejected) with the auditor's *agreement rate* (whether it should have been) — noting that an LLM audit measures model disagreement under asymmetric information, and ground truth is reserved for the blinded human lane (§ Results). High restraint + low audit agreement would be a warning, not a success.

- **System contrasts against baselines.** The same candidate stream is run through an ungated screener, debate-only, no-falsifiable-contract/lock, and no-overlap-penalty variants — the design goal being to show the full pipeline admits *fewer* ideas **and** (once arm-specific audits exist) that its rejections are better justified — not merely that gates reduce the count. The pilot ran the admission-rate half only (§ Results).

## 7 Experiment Plan (Pre-Registered)

This plan is itself pre-registered: it commits the evaluation **before** the forward data is collected, consistent with the framework's own discipline. Edit only by appending dated amendments.

### Hypotheses

- **H1 (restraint):** the gated, forward-registered pipeline admits a **low fraction** of the candidates it *evaluates* (and a lower fraction of those it *sources*). Pre-committed threshold: admission rate among evaluated finalists $<$ **10%** over the study window. (The prose "low" and the numeric threshold are deliberately the same single-digit target — no looser fallback.)

- **H2 (each discipline contributes):** ablating any one of {deterministic gate, bear-case-first, overlap penalty} **raises** the admission rate vs the full pipeline. *Supported for an ablation if* its admission rate exceeds the full pipeline's by $\geq$ **25% relative OR** $\geq$ **5 percentage points, whichever is larger** (the max guards against a tiny relative jump on a tiny base, and against a large relative jump that is still trivial in absolute terms).

- **H3 (portability):** the disciplines transfer to a second domain with only `bottleneck_map.yml` changed. *Supported if* (a) the second domain's admission rate stays **within 2**$\times$ the first domain's rate, **and** (b) $\geq$ **70%** of the second domain's rejections map to the **same gate / rejection-category taxonomy** used in the first (tradeability, liquidity, low purity, overlap, valuation, crowding, unverifiable-trigger); **and** (c) the second-domain run uses the *same engine code and prompt templates*, with changes limited to `bottleneck_map.yml`, the domain-specific source lists, and the benchmark config — otherwise "portability" could be satisfied by quietly modifying the system. "Comparable" is thus defined on rate, the failure-mode taxonomy, and a no-source-change constraint, not asserted qualitatively.

- **H4 (forward, reported honestly):** over the forward window, benchmark-relative excess return and information ratio on conservative shadow fills are reported with confidence intervals. *No threshold is pre-set for H4 with a small sample* — it is descriptive, not a success gate.
- **H5 (rejection *quality*, not just quantity):** restraint is only evidence of discipline if the rejected ideas were actually bad. We therefore audit rejections: an independent reviewer (a held-out LLM auditor and/or a human), given only the *bull* case and the decision — **not** the engine's bear case — labels each sampled rejection JUSTIFIED or a FALSE REJECTION and assigns a category (low-purity / duplicate-overlap / valuation-or-crowding / commodity-cycle / pre-revenue-or-hype / unverifiable-or-untradeable). *Supported if* **rejection precision $\geq$ 0.80** ($\geq$ 80% of sampled rejections judged justified) **and** the false-rejection rate is $\leq$ **10%**. This directly answers "a system can reject 95% of ideas by being arbitrarily conservative" — H1 measures *how much* is rejected, H5 measures *whether it should have been.*

## Design

- **Domains:** the AI-supply-chain worked example, then 1–2 additional bottleneck domains as configs.
- **Horizon:** $\geq$ 6 months of forward observation per domain before any performance claim; discipline metrics reported from the first runs.
- **Unit of analysis:** a candidate (sourced $\rightarrow$ gated $\rightarrow$ evaluated $\rightarrow$ admitted/not) and a position (admitted $\rightarrow$ forward outcome).
- **Primary metrics:** sourced/gated/evaluated/admitted counts; admission rate; no-action rate; source-tier distribution; overlap-penalty distribution; ledger-integrity (% admissions with a valid hash, chained to the prior entry, registered before the position); **rejection precision** and **false-rejection rate** from the H5 audit.
- **Secondary (forward):** excess return vs the theme benchmark; information ratio; active drawdown; gross/net thematic beta; turnover; shadow-fill slippage.
- **Ablations and baselines:** rerun the same candidate stream against (a) **ungated LLM screener** (no deterministic gate); (b) **debate-only** (bull/bear with no pre-registration lock); (c) **no Forward-QPOP lock** (same evaluation, hypotheses not content-hashed before the window); (d) **no overlap penalty**; and, where feasible, (e) a **human/manual thematic shortlist** for the same domain. Report each one's admission rate **and** its rejection precision — the point is to show the full pipeline admits *fewer* ideas *and* that its rejections are better justified, not merely that gates reduce count.

## Decision Rules (Pre-Committed)

- H1 is **supported** if the evaluated-finalist admission rate is $<$ **10%** over the window.
- H2 is **supported** (per ablation) if that ablation's admission rate exceeds the full pipeline's by $\geq$ **25% relative or $\geq$ 5 percentage points, whichever is larger**.
- H3 is **supported** if the second-domain admission rate is **within 2$\times$** the first-domain rate, **and $\geq$ 70%** of its rejections map to the shared rejection-category taxonomy, **and** the second-domain run uses the *same engine code and prompt templates*, with changes limited to `bottleneck_map.yml`, the domain-specific source lists, and the benchmark config (clause (c) — otherwise "portability" could be satisfied by quietly modifying the system).
- H5 is **supported** if **rejection precision $\geq$ 0.80** and the **false-rejection rate $\leq$ 10%** on the audited sample. *Dated note (2026-07, recorded rather than silently rewritten): when both clauses are computed on the same binary-labeled audit sample, the false-rejection rate equals one minus precision, so the two clauses are mathematically redundant and the $\leq$ 10% clause effectively binds at precision $\geq$ 0.90. Both clauses fail on the documented sample (§ Results). The prospective H5 window will use a single explicit criterion committed in a dated amendment before its data accrue.*
- A position's outcome is recorded **Supported / Weakened / Falsified** strictly by its pre-registered exit triggers — never by an after-the-fact narrative. Because a thesis is judged against *several* triggers over time, outcome decisions are **designed to use a sequential test with explicit Type-I error control** (an anytime-valid / e-value formulation from the safe-testing literature (Ramdas et al., 2023; Shafer, 2021; Vovk and Wang, 2021); specified in the pre-registered plan; an *experimental* implementation is ledger-integrated in the released package, not used in the pilot decisions and not yet delivering the guarantee[2]), the design intent being that repeated looks at *one* hypothesis's registered triggers not inflate false "Falsified" calls for that hypothesis — per-hypothesis anytime-valid control, applied *prospectively* once the implementation matches the rule. Book-wide (across-hypothesis) multiplicity control — the factor-zoo analogue — is *not* provided and is declared future work.

[2]The e-value / anytime-valid module in the released package (`forward_qpop.evalue`) is an *experimental implementation of a theoretically anytime-valid per-hypothesis rule* — it does *not yet deliver* that guarantee, and it is not pilot evidence. The rule: a betting-martingale e-process per registered trigger, an arithmetic-average combiner by default (dependence-safe per Vovk and Wang (2021); the product combiner is valid only under a genuine independence assumption) and Ville-threshold decisions, covered by 18 module tests (including a Monte-Carlo Type-I check under optional stopping) plus 17 ledger-integration tests. The rule's guarantee, once the implementation matches it, is **per-hypothesis**: anytime-valid control for that hypothesis's registered triggers under a stated conditional null ($\Pr(\text{fire}_t = 1 \mid \text{past}) \leq p_0$), assuming non-overlapping observation periods — a persistent fired state must not be re-counted as fresh evidence; no across-hypothesis (book-wide) multiplicity control is provided under any configuration. Two known implementation defects currently break the stated rule — the mixture initializes each trigger's e-process only when that trigger first reports (changing mixture membership/weights after observations, where the rule requires all registered triggers initialized at 1 with fixed weights) and trigger values are coerced with `bool()` rather than strictly type-checked — and must be closed, with regression tests, before the module drives a live outcome decision. It is wired into the ledger: an admission entry *can* carry a frozen `evalue` commitment (`p0`/`p1`/combine, hashed with the entry), belief-update entries may carry `trigger_checks` observations, and `forward-qpop evalue <ledger.jsonl>` runs the sequential test with its state persisted in a sidecar (exactly-once resumption; never mutating the ledger). The reporting-time significance level $\alpha$ is a CLI argument, *not* part of the hashed commitment; any operational decision threshold must be committed in a separately dated policy. It was *not* used in the pilot outcome decisions — the pilot predates it — and as of this revision no ledger admission carries a live `evalue` commitment (a direct consequence of the zero-admit streak since the integration landed); the first future admission can freeze its commitment at registration.

### Implementation Details (Fixed in Advance, for Reproducibility)

Recorded here so the forward log is reproducible and leakage-resistant; any change is a dated, additive amendment.

- **Model versions & prompts:** the exact model identifier (and version/date) for each tier (SOURCE / first-pass EVALUATE / adjudication) is logged per run; prompt templates are committed in the repo and referenced by commit hash. A model-version change is an amendment, not a silent swap. *Disclosure (v2):* that per-run log is private, and the *executed* pinning mechanism is per-run **SHA-256 hashes of the rendered prompt artifacts** in the private provenance record (the registered "referenced by commit hash" phrasing above describes the private implementation's versioning, corrected here as a dated amendment) — the public `src/prompts.md` releases sanitized reference templates (the H2 arm prompts verbatim), not the production bundle; the public paper names only the cost tiers (cheap / mid / expensive). A sanitized exact-version run table (identifiers, dates, invocation counts, token/cost totals, with missing fields exposed as missing) is not yet released, so the cost-aware-cascade contribution should be read as *architectural* (the tier allocation and its asymmetry) rather than a publicly reconstructible per-run cost accounting.
- **Source timestamps:** every evidence item carries the publication date of the source and the retrieval date; an admission may rely only on evidence dated **on or before** the admission date.
- **Price source:** a single declared price vendor (e.g. daily adjusted closes) with corporate actions (splits/dividends/spin-offs) applied via the vendor's adjusted series; the vendor is named in the run config.
- **Paper-trading timestamp & fill rule:** admissions take effect at the **next session's opening print** after the content hash is registered (never the same bar as the signal). Forward track-record metrics use a **conservative shadow fill** = opening print + **assumed slippage (default 10 bps)**; broker paper fills validate automation only and are not the metric of record.
- **Transaction cost / slippage:** the 10 bps slippage assumption plus declared commission (default $0) are applied to every shadow fill; turnover is reported so costs are auditable.
- **Corporate actions:** handled via the adjusted price series; admissions/exits triggered by a corporate action (e.g. acquisition close) are logged as such, not as thesis outcomes.
- **Post-admission news:** the system **may** read post-admission news, but it may only *fire a pre-registered exit trigger or open a new dated belief-update entry* — it may **not** retroactively edit the original hypothesis or its triggers (the content hash makes such an edit detectable).

### Integrity Controls

- Every admission carries a hash-chained QPOP entry (each entry binds the prior entry's hash) registered before the position — tampering with any past entry breaks the chain.
- The ledger is append-only; closed entries are immutable.
- No metric is changed after seeing results; amendments are dated and additive.
- Forward results are **evaluated** on conservative shadow fills, not optimistic fills.

- **Structural-validity reporting (Ye et al., 2026):** any forward performance number is reported only alongside its structural-validity checklist — temporal integrity (admission strictly precedes the evaluation bar), frictions (slippage + costs applied), robustness (across the accumulated ledger, not a cherry-picked window), calibration (admitted confidence vs realized outcome), and execution (shadow-fill rule). A return without this checklist is not reported as evidence.

### Threats to Validity (Declared in Advance)

Small sample / wide intervals early; LLM nondeterminism across model versions; survivorship in which domains get built; the admission-rate metric is *necessary but not sufficient* for forward predictive performance. The paper reports these rather than hiding them.

## 8 Worked Example: AI Supply Chain

We instantiate the framework on the AI-accelerator supply chain. One discovery round, end to end (the batch shown in full in Figure 2): about fifty candidate cards are sourced across twelve lenses; the persistent seen-set leaves roughly two dozen genuinely new symbols; the deterministic gate passes twenty (foreign-only listings and sub-liquidity names are rejected mechanically); a cheap triage screens the survivors for obvious low-purity names, and the bear-case-first evaluation plus adjudication admit *none*. The modal outcome is no action. **And this is not a single-batch artifact:** over the forward window **2026-06-08 through 2026-07-06**, the run ledger records **9 discovery rounds — 318 candidate cards sourced, 110 reaching gate-pass evaluation — with *no* new admission** (a ledger-derived aggregate, frozen at a dated evidence cutoff: the script-generated artifact committed 2026-07-09, source commit `1744608`, sha256 `bb0a9507...`; the window end is the cutoff, not a moving build-time regeneration. The 2026-06-07 build-out day, with its 3 admits, falls outside this forward window). This is a count of *ledger-proper* discovery rounds; the round-by-round `DISCOVERY_QUEUE` log runs a longer consecutive-zero-admit streak, and we keep the two denominators distinct. Restraint is recorded round by round, not in one cautious batch. Across the pilot build-out (including earlier sessions, not only this 0-admit round), we show three representative dispositions: **admitted** — a supplier-concentrated critical-material chokepoint, sized at half its unconstrained satellite weight by a separate concentration cap and admitted only with a full set of pre-registered, primary-source exit triggers (a bottleneck-break falsifier, an execution/margin trigger, and a crowding-reversal trim) — restraint expressed as sizing *and* as falsifiability, not a price target; **rejected** — an integrated energy major proposed for a "helium chokepoint" whose helium revenue is under one percent of the company (a real chokepoint the *ticker* does not capture); and **overlap-penalized** — a high-purity optical pure-play routed to a replace-not-stack decision because its node sleeve already held the incumbent it would duplicate. A full sanitized walk-through is the released `examples/ai_supply_chain/` domain. Separately, a diversifier sweep on **2026-06-14** produced one further admission on the same gated, pre-registered terms — a de-rated, low-AI-beta defense-industrial chokepoint taken at half size specifically to reduce the book's single-factor concentration — evidence that the pipeline *does* admit, across domains, when a candidate earns it. To reconcile the counts in one place: the build-out era admitted **4** names in total (3 on the 2026-06-07 build-out day + this 2026-06-14 diversifier, which carries its own pre-registered hypothesis entry and is not a discovery-funnel round), against **0** admissions across the 9 counted forward discovery rounds of 2026-06-08 through 2026-07-06.

# 9 Results

**Pilot metrics — not a performance claim, and not confirmatory.** This paper validates *process discipline*, not investment profitability; investment validity requires the H4 forward window. The numbers below are **pilot / initial discipline metrics** from early candidate batches: they *motivate* the registered evaluation and are labeled as such, *not* treated as confirmatory evidence. Where a threshold (H1's `<10%`, H5's $\geq$`0.80`) was not locked strictly before these specific batches were observed, the batch number is a pilot reading against that threshold, and the pre-registered plan governs only the forward study window. Forward returns are not yet observed.

Table 3 states, for each hypothesis, what its current evidence rests on, whether its threshold preceded the data, and the analysis type — so a skeptical reader can tell which claims are exploratory and which are genuinely forward-registered without consulting private notes.

| **Hyp.** | **Evidence set (dates)** | **Threshold locked before data?** | **Type** | **Current reading / next read** |
|---|---|---|---|---|
| H1 | 2026-06-18 batch; ledger window 06-08..07-06 (frozen artifact) | No — plan registered contemporaneously with early batches | pilot | 0/20 finalists; 0 admits over 9 rounds; confirmatory read: forward window |
| H2 | 38-card batch (3 pilot arms, single-run); +2 arms 2026-07-07, two runs each (dated retrospective amendment) | No | exploratory contrasts | inconclusive — not tested as registered; matched ablation is future work |
| H3 | pharma (v1 era); uranium 2026-07-07; launch/satellite (v1 era) | Clauses registered before the uranium run; pharma contemporaneous | retrospective amendment + exploratory | not supported as registered (no single domain meets (a)+(b)+(c)); joint single-domain test pending |
| H4 | forward paper cohort, live since mid-June 2026 | Yes (descriptive by design; no threshold) | prospective | accruing; first read ~2026-12 |
| H5 | $n$=40 seeded draw 2026-07-07 from a 48-case population; pilot $n$=14 undocumented | Gate registered 2026-06-29 (v1); the audited rejections largely predate it; the pilot predates the gate entirely | pilot | both clauses fail (0.775 agreement; 22.5% flag rate); human lane pending; rejection quality unresolved |

Table 3: Evidence status by hypothesis. "Type" distinguishes pilot readings, retrospective amendments, and prospective (forward-registered) claims; none of the pilot batches is a confirmatory test. Registration artifacts: the experiment plan ships in the public repository; amendments are dated in the text.

## 9.1 The Funnel, on a Real Batch

A discovery session (2026-06-18, AI-supply-chain domain) over under-covered chokepoint layers, sourced across 12 lenses (per-node, gaps, social, policy, and literature), produced the cascade in Figure 2: 51 candidate cards collapsed to 24 unique new symbols after seen-set dedup; 20 passed the deterministic gate (tradeability + liquidity + overlap; foreign-OTC and sub-threshold-liquidity names rejected mechanically); and the bear-case-written-before-recommendation evaluation plus adjudication admitted **0** this session — the tenth consecutive zero-admit round (prior build-out sessions admitted 3, all at small satellite).

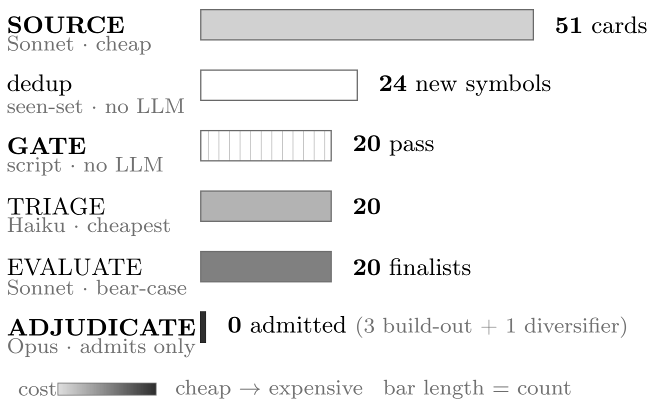


Figure 2: The discovery cascade on a real batch (2026-06-18, AI-supply-chain). Bar *length* is the candidate count surviving each stage; bar *fill* is the model cost tier (white = deterministic, no LLM). The dedup step removes duplicates, not ideas; the expensive model judges only the survivors and admitted **none** this session — this is rejection, not throughput, and "no action" is the modal outcome.

Admission rate among evaluated finalists **this session: 0/20** (rule-of-three 95% upper bound ≈ 15% on so small a sample; the unique-new-symbol admission rate was **0/24**); across the forward window (2026-06-08 through 2026-07-06) the ledger aggregates to **0 admits over 9 discovery rounds and 110 gate-pass evaluations**, and the trailing admission rate across the full build-out (including the 3 earlier build-out admits) is **low single digits** — a pilot reading consistent with H1 (`< 10%`), to be confirmed over the forward window with proper intervals. These consecutive 0-admit rounds were *not* a failure of sourcing; in the featured batch the gate-passers were diversified mega-caps (chokepoint = a tiny segment) or pre-revenue, and the genuinely pure vehicles surfaced were foreign-OTC and failed the US-tradeability gate. "No action" is the modal outcome.

## 9.2 Rejection-Quality Audit (H5)

The central objection to a low admission rate is that *a system can reject 95% of ideas by being arbitrarily conservative.* So a **held-out, bull-only LLM auditor** (a different model instance — an *adversarial diagnostic*, not a claim of model-independent audit; human review is planned for the forward study, not yet done) was shown only the steelmanned **bull** case and the decision — **not** the engine's bear case — and asked to label each sampled rejection JUSTIFIED or FALSE REJECTION with a category.

**The documented sample supersedes the pilot's.** The pilot's 14-rejection membership and its sampling method were never recorded; the expansion therefore draws from a fully enumerated population — all 48 non-admitted candidates of the batch-era rounds (mechanical bull-only extraction,

| **Metric** | **Expanded ($n$=40, seeded)** | **Pilot ($n$=14, undoc.)** |
|---|---|---|
| Sampled rejections audited | 40 | 14 |
| **Bull-only auditor agreement** (primary) | **0.775 (31/40)** | 0.929 (13/14) |
| Wilson 95% CI (naïve, descriptive) | $[0.625, 0.877]$ | — (non-inferential) |
| Bull-only flag / disagreement rate | 22.5% (9/40) | 7.1% (1/14) |
| Full-record LLM uphold among flagged (secondary) | 9/9 | 1/1 |

Table 4: Rejection-quality audit, reported as *model disagreement under asymmetric information*, not ground truth: the bull-only auditor sees only the bull case; the full-record adjudicator (a related model) sees the full record, so overturned flags do not establish that the engine's rejections were correct. Denominators: agreement and flag rates are over all audited rejections ($n$); the uphold rate is over flagged cases only. The documented $n$=40 seeded sample is primary; the pilot $n$=14 (membership and sampling method never recorded) is retained only as a clearly-labeled pilot reading, its interval suppressed as non-inferential. The H5 gate (registered 2026-06-29, after most of the audited rejections existed — see Table 3) required agreement (registered as "raw precision") $\geq 0.80$ *and* a flag rate $\leq 10\%$ — **both clauses fail** on the documented sample (0.775; 22.5%); on a single binary-labeled sample the two clauses are redundant (flag rate = 1− agreement), effectively binding at 0.90 (see the dated note in § Decision Rules).

bear-case exclusion enforced by test) — at a fixed seed, round-stratified, with a content-hash-verifiable manifest ($n$=40). Five fresh runs of a held-out bull-only auditor (eight records each; fresh contexts within one model family, *not* statistically independent instances) labeled each rejection JUSTIFIED or FALSE REJECTION; every flag was escalated to an expensive-model, capital-at-risk adjudication on the full record (a no-web retrospective — a dated deviation from the pilot's contemporaneous adjudication, to avoid look-ahead).

**On the documented sample the bull-only auditor agreed with** 31/40 (0.775) **of the engine's rejections and flagged** 9/40 (22.5%)**, so *both* H5 gate clauses fail (**$\geq 0.80$ **agreement;** $\leq 10\%$ **flag rate) — we record the gate failure as a result, not a footnote.** Because the gate was registered in v1 (2026-06-29) while the audited rejections largely predate it and the pilot predates it entirely, this batch is a pilot reading against the criterion, not a confirmatory test of a criterion that preceded its data. The pilot's 0.93 was the optimistic read of a small, undocumented sample: the expanded point estimate lies *inside* the pilot's own interval, so nothing is contradicted, but the $n$=14 headline did not survive a 2.86× larger, documented sample (the flag count rose 9×, from one to nine). The Wilson interval is itself naïve and descriptive: the 40 labels come from five eight-record runs in one model family over an 83% draw of a 48-case population, so they are neither independent nor an infinite-population sample.

The full-record adjudication layer is an *escalation diagnostic*, not independent validation. The full-record adjudicator overrode all **nine** bull-only flags (9/9): in each case it cited a load-bearing fact from the full record that the bull-only auditor — shown only the steelmanned bull case — could not see. Because auditor and adjudicator are related models judging under deliberately asymmetric information, a full-record model overriding bull-only flags is *expected*; it measures where cheap bull-biased review and full-record review disagree, and it does *not* establish that the engine's original rejections were correct. The flags split *evenly* across the funnel's cheap-drop and full-eval stages (5/4 against a 20/20 split), so the auditor/engine disagreement is not a sample-composition artifact of one stage. One illustrative flag: the auditor judged a primary-sourced design win admissible where the engine had said "watchlist"; the full-record pass surfaced that the headline win is **shared across a 14-member supplier alliance that already includes a held name** — one of fourteen substitutable suppliers, not a chokepoint owner — and the flag was overridden. The methodological point is the cost asymmetry: a cheap, bull-biased audit *should* occasionally push to admit, and the expensive full-record layer exists to catch exactly that — the same cheap-screen → expensive-adjudicate asymmetry that governs admissions — not a claim that the overrides are ground truth. **Rejection quality therefore remains unresolved** pending the human lane, whose protocol is pre-specified in §"What Is Still Pending."

### 9.3 Exploratory System Contrasts (H2): Admission Under Weakened Discipline Configurations

A separate **38-candidate batch** (distinct from the funnel batch in Figure 2) was run through baseline arms that each *weaken or remove* a discipline, and their admission rates compared to the full pipeline. To keep the comparison honest about what is and is not matched: the candidate cards are identical across all arms, and the *single-pass* arms share one model identifier and temperature; the full-pipeline comparator, however, is the production staged cascade (multiple model tiers, gate, seen-set, triage, adjudication), so the full-vs-arm contrast changes several components at once and is a *system contrast*, not a matched one-factor ablation. Every arm may choose ADMIT / WATCHLIST / REJECT under the same capital-at-risk context. The three original arms (ungated, bear-case-off, overlap-off) were run in the pilot as single runs; the two additional arms (**debate-only** and **no falsifiable-contract / forward-lock requirement**) were then run on the *same* preserved 38-card batch under the identical single-pass protocol, with **two fresh runs each** (dated amendment, 2026-07-07, recorded in `research/docs/RESULTS_V2_WORKING.md`: a retrospective run; two runs of one model are not independent replicates). All arm prompts are released (`src/prompts.md` §8) so the baselines can be inspected; they are explicitly *allowed to reject*, not strawmen forced to admit.

| Arm | Admitted | Rate | Repl. agr. |
|---|---|---|---|
| **Full pipeline** (gate + seen-set + triage + bear-case-first + overlap) | **0 / 38** | **0%** | — |
| Debate-only (bull/bear exchange, no pre-registration lock) | 15/38, 13/38 | ∼37% | 89% |
| No falsifiable-contract / forward-lock requirement (same evaluation; no lockable hypothesis contract, no content-hash pre-window) | 14/38, 17/38 | ∼41% | 82% |
| − overlap penalty (judge each name on its own, ignore duplication with the held book) | 25 / 38 | 66% | — |
| − bear-case-first (recommend from the bull thesis, no written bear case) | 28 / 38 | 74% | — |
| **Ungated LLM screener** (same cards + admit/watchlist/reject, but no gate, purity bar, overlap penalty, or bear-case requirement) | **38 / 38** | **100%** | — |

Table 5: Exploratory system contrasts on the 38-candidate batch, ordered by admission rate *after the fact*. The two arms added after v1 (debate-only, no-contract/lock) report two fresh runs each with per-card agreement; the pilot arms are single-run. The arms are not matched one-factor ablations (see text).

*Audit boundary of the baselines.* The ungated baseline was **allowed to reject** — it is not a strawman forced to admit. What a reader can inspect from the public release: the *literal arm prompts* (`src/prompts.md`, released verbatim, including the ungated arm's instruction set) and the aggregate per-arm decision counts reported here — enough to judge whether the ungated arm's *instructions* were admission-biased. What a reader cannot do: re-run or re-score the arms, because the empirical 38-card set, the per-card arm outputs, and the run manifest remain in the private research repository (only a schema-conformant synthetic card sample is public). The fairness of the 100% is therefore *inspectable at the prompt level, not independently reproducible.*

In this batch and prompt setting the contrasts are large: the **ungated screener admitted every candidate** (the "over-admits by construction" failure made concrete); every weakened-configuration arm admits far more than the full pipeline (each difference exceeds the absolute limb of H2's threshold, $\geq$ 5 pp; with a zero full-pipeline rate the $\geq$ 25%-relative limb is undefined, a zero-denominator convention we record here as a dated amendment); and the full stack sits at 0%. **H2 is nevertheless inconclusive — not tested as registered.** The registered design promises each arm's admission rate *and rejection precision*; no arm-specific rejection-quality audit was run, so the rejection-precision half is unmeasured. The arms are not matched one-factor ablations: several remove more than one discipline at once, the single-pass arms and the staged production comparator differ in more than the named discipline, and only the two new arms have a second run. What the contrasts show is that large admission-rate differences accompany the removal of disciplines; they do not identify any single component's causal contribution. We read the 100% honestly: an ungated screener that is free to reject but admits everything is exhibiting exactly the structural bias-to-admit the framework exists to counter — the informative quantity is the *change* in behavior across configurations, not that any single arm always admits.

**With the two new arms the observed rates, sorted after the fact, span** 100% $\rightarrow$ 74% $\rightarrow$ 66% $\rightarrow$ ~41% $\rightarrow$ ~37% $\rightarrow$ 0% (ungated, −bear-case, −overlap, no-contract/lock, debate-only, full pipeline). The ordering is descriptive — these are different, non-nested interventions, so it is not a dose-response curve and we draw no monotonicity or saturation claim from it. The new arms land *between* the full pipeline and the single-discipline ablations: the most disciplined single-pass prompt — bull/bear reasoning and purity intact, without the falsifiable-contract requirement — still admits **roughly four in ten**. The residual gap to 0% coincides with the components the single-pass arms lack — the deterministic gate, the cheap-triage escalation asymmetry, the capital-at-risk adjudication, and the lockable-contract bar as practiced — but these contrasts cannot identify which of those components, if any, is causal; that attribution awaits a matched, nested ablation (future work). Consistently, the new arms' *consensus-admit core* (the names both runs admit) closely tracks the production funnel's WATCHLIST tier — one such consensus name later earned a pre-registered flip contract — so prompt-level discipline finds the same candidates the staged funnel does, but the funnel converts "interesting" into "watch with a dated trigger" rather than "commit capital now." Both new arms' admission rates exceed the full pipeline's by more than the absolute $\geq$ 5 pp limb, with high per-card agreement between the two runs (89%, 82%).

> A reviewer's exact objection — "a system can reject 95% of ideas by being arbitrarily conservative" — is *partially* answered: the contrasts show the restraint is *configuration-dependent* — admission jumps when the discipline configuration is weakened — rather than blanket conservatism baked into the underlying model, though which specific component drives it is not identified. Whether the rejections are *justified* is not yet established — the bull-only audit agreed with 31/40 and the full-record adjudicator overrode all nine disagreements, an intra-model-family diagnostic under asymmetric information, not ground truth; the human lane is the outstanding test.

**Over-admission rate as a reusable benchmark.** We name the quantity these contrasts measure — the **reference-relative over-admission rate**. For a fixed candidate set $C$ evaluated by an arm

$A$ against a reference discipline $R$,

$$\mathrm{rOAR}(A, R; C) = \frac{|\,\mathrm{admits}(A) \setminus \mathrm{admits}(R)\,|}{|C|}, \qquad \mathrm{rOMR}(A, R; C) = \frac{|\,\mathrm{admits}(R) \setminus \mathrm{admits}(A)\,|}{|C|},$$

with the denominator pinned to the *fixed candidate-set size* $|C|$ — never the arm's own admit count, which would let an arm drive the metric toward zero by admitting almost nothing — and watchlist treated as non-admit. Both directions are defined because the reference is a *reference*, not ground truth: a full-pipeline rejection can itself be wrong, so rOAR measures disagreement with the reference discipline, and "true" over-admission is reserved for candidate sets with independent labels or settled forward outcomes. In this pilot the reference admitted zero of the 38-card batch, so rOMR $= 0$ *by construction* — an artifact of the zero-admit reference, not evidence the reference is infallible. Operationally: the ungated screener's rOAR against the full discipline is 100% on this batch, and every weakened-configuration arm moved rOAR by more than H2's absolute $\geq 5$ pp limb. A meaningful benchmark pairs rOAR with a *rejection-quality* audit of the reference itself (here: bull-only agreement 31/40, unresolved pending the human lane): driving rOAR down matters only if the reference's rejections are defensible rather than blanket conservatism. We report a single-domain, reference-relative pilot; a multi-domain OAR benchmark — constructed candidate sets with known should-reject labels, blinded auditors, and a shared task format — is future work, and an explicit invitation to build on this release.

### 9.4 Portability (H3): Four Probes Across Domains and Substrates

Table 6 summarizes the four probes before the narratives; the H3 verdict follows from it.

| Domain | Substrate | Code/config | Finalist outcome | Clauses probed |
|---|---|---|---|---|
| AI supply chain (ref.) | physical | production engine | 0/20 admitted (featured batch) | — |
| Pharma / GLP-1 | physical | node list changed | 0/7 admitted; 2 watchlist, 5 reject | (a), (b) |
| Uranium fuel cycle | physical | one config file, zero code edits | 1/4 admit-class (probationary, recorded only) | (c) |
| Launch / satellite | physical | unmodified | 0 admitted | taxonomy only |
| Proprietary-data | non-physical | design-stage | no live run | none (App. A) |

Table 6: The portability probes. No single second domain satisfies H3's clauses (a)+(b)+(c) jointly — (a)/(b) read on pharma, (c) on uranium — so H3 is **not supported as registered**; the clause-level readings are exploratory successes on different domains.

To probe H3 we ran the *same* funnel on a deliberately distant domain — **pharmaceutical injectables / GLP-1 drug delivery** — changing only the node list. The funnel narrowed the same way: **27 candidate cards sourced** → 17 with a US-tradeable symbol (10 were real chokepoints whose only vehicle is foreign / private, the same "no clean domestic vehicle" failure seen in the AI domain) → deterministic gate **7 pass / 10 fail** (foreign-OTC ADRs + sub-liquidity) → triage + bear-case + adjudication → **0 admit, 2 watchlist, 5 reject**.

The disciplines transferred. The engine *located* the domain's canonical chokepoint on the first pass — a coated-elastomer closure near-monopoly whose components are named in FDA/BLA filings, so substitution requires multi-year change-control revalidation (the pharma analogue of the substrate-film chokepoint in the AI domain) — and scored it high on both bottleneck strength and purity. It **rejected the five conglomerate look-alikes** on *low exposure purity* (the dominant rejection category in both domains; the chokepoint is a small segment of a diversified parent). And the **entry discipline fired independently of the chokepoint-quality test**: the bear-case pass flagged the pure chokepoint for admission, but adjudication *downgraded it to watchlist* on exactly the valuation/crowding logic the first-domain book applied to its own real-but-priced-in names — a stretched multiple with a thesis that reduces to "the multiple holds."

Measured against the pre-registered H3 criteria, **H3 is not supported as registered**: the registration requires a single second-domain run to satisfy clauses (a), (b), and (c) jointly, and no single domain does. The clause-level readings are exploratory successes on *different* domains. **(a)** reads on the pharma run: its evaluated-finalist admission rate (0/7) matches the first domain's zero — though "within 2×" of a zero rate is degenerate as a ratio test (it admits only exactly zero), and a dated amendment will replace it with an absolute non-inferiority criterion before the prospective window. **(b)** also reads on pharma: every recorded rejection was assigned a category from the registered taxonomy (low-purity-conglomerate, foreign / no domestic vehicle, illiquid, priced-in-watchlist); the registration did not pin which rejection stages enter the denominator, so we state this as a complete-assignment observation rather than the previous "100%" percentage. **(c)** reads on the uranium run — but uranium's own evaluated-finalist outcome was **1 admit-class flag of 4 evaluated** (upheld at probationary half-size, recorded only), a rate the zero-anchored clause (a) cannot score, so uranium does not satisfy (a). A single-domain joint test (all three clauses on one config-only domain) is pending. On clause (c) itself: since v1, the `domain` profile has been threaded through the SOURCE / EVALUATE templates — byte-identical golden tests pin the live domain's prompt bytes — so a full funnel on a genuinely new domain, the **uranium fuel cycle** (named as a candidate in v1's Discussion), required **one new configuration file and zero prompt-construction code changes** (the change set is added-files-only; no generated prompt contains a first-domain string). The run narrowed the same way: 20 cards sourced across five lenses → 10 unique symbols → deterministic gate 8 pass / 2 mechanical rejections (one foreign-OTC sub-liquidity, one min-price) → cheap triage 4 pursue / 3 drop → bear-case-first evaluation 1 admit-class flag / 2 watchlist / 1 reject → expensive adjudication **upheld the single flag at probationary half-size** with five dated exit triggers (a recorded disposition only — the demo book is unfunded). The full rejection taxonomy, the source-tier and citation-verification discipline, and the escalation asymmetry in *both* directions transferred unprompted, and this is the framework's first admit-class outcome in any second-domain probe — restraint is not blanket conservatism. *Release boundary:* the domain-templating, gate, and scoring engine that performs this swap is part of the private research book, so a reader **cannot** re-run the config-only swap from the public repository; the public artifacts are the ledger, hash chain, timestamp anchor, e-value module, schemas, and worked example domains (see § Reproducibility).

**A third domain, and a new rejection pattern.** As a further portability probe we ran the same funnel on a third, distant domain — the launch / satellite-internet supply chain, grounded in the recent S-1 of its dominant operator. The engine transferred unmodified and again produced a decisive negative (**0 admit**). Two of the prior domains' patterns recurred and a new one emerged. **(i)** The rejection taxonomy held: low-purity conglomerates, foreign / no-US-vehicle chokepoints (one UK-listed and several Asia-listed), and priced-for-perfection cyclicals trading near all-time highs were all rejected on the same gates. **(ii)** The engine caught a class of error a naive "supplier-of-X" screen does not — it flagged apparent suppliers that are in fact *competitors* or *spectrum counterparties* of the target rather than input vendors. **(iii)** A substitution mode appeared that the prior domains never exhibited: the customer *vertically integrating the chokepoint away.* The target actively engineers out its own named inputs (in-sourcing propellant pressurization, primary structures, and user-terminal silicon), so a "supplier" thesis must additionally survive the customer building the part itself. The engine absorbed this without modification, adding *"can the customer in-source this input?"* to the bear case alongside the second-source, new-technology, and policy-substitution tests. A third distant domain therefore reinforces the *method-level* portability claim — the disciplines and the rejection taxonomy transfer — and the new bypass mode shows the bear case is extensible to domain-specific substitution paths rather than fixed to the first domain's.

**A fourth probe, across *substrates*.** As a further portability probe we extended the same disciplines to a *non-physical* substrate — proprietary-data and network monopolies. That probe is design-stage only, so we report it separately in Appendix A as an exploratory, hand-selected case series: the disciplines transferred (the same purity and over-payment gates separated genuine data monopolies from contestable look-alikes), and it *generates* — but does not test — a substrate-specific hypothesis about the *entry* rule.

### 9.5 What Is Still Pending (Declared)

**Delivered since the v1 submission** (all dated 2026-07 in the released working record, `RESULTS_V2_WORKING.md`): both remaining ablation arms (debate-only, no-contract/lock — § H2 above); config-only domain portability (the uranium funnel — § H3, clause (c)); the expanded rejection-quality audit ($n$=40, on which *both* gate clauses failed (a criterion registered 2026-06-29, after most of the audited rejections existed), reported as such — § H5); the **e-value / anytime-valid** module, now *ledger-integrated* (a frozen per-admission `evalue` commitment hashed with the entry, plus a `forward-qpop evalue` CLI that runs the sequential test over the ledger; 18 module + 17 integration tests — an experimental implementation of a per-hypothesis anytime-valid rule that does *not yet deliver* the guarantee (two known defects tracked; § Decision Rules footnote), not used in the pilot, and no live admission carries a commitment yet); and an external **OpenTimestamps** anchor (opt-in / manual by design, 11 hermetic tests plus a dispatch-gated live CI job). Still pending:

- **Forward outcomes** (H4) — accrue over the forward window; reported only with the structural-validity checklist. The forward book is *deployed:* it runs as a daily paper-trading cohort (live since mid-June 2026), which validates the automation end-to-end while the shadow-fill rule remains the metric of record — a deployment-status fact, not a performance claim. Accrual toward the pre-registered ∼100-observation December read is ongoing, and a drift-band monitoring gate was pre-registered 2026-07-09 with thresholds frozen before a fittable history existed (alert-only until its arming criterion). First read-out ∼2026-12.

- **Human-audit lane** for H5 — *redesigned per review, superseding the earlier bull-only packet* (a human reviewer shown only the bull case would still measure disagreement under asymmetric information, not ground truth): two human reviewers receive a neutral, dated *full-record* packet, blind to all LLM labels and final pipeline decisions, with source verification allowed, a pre-specified third-reviewer tie rule, and inter-rater agreement reported. Until it runs, rejection quality remains unresolved.

- **Forward-scoring registry read-out** — the engine-level cohort is seeded and frozen forward (the first cohort is recorded); it is provisional until it accrues sufficient name–horizon observations, with the first read-out roughly two quarters out, and is likewise reported only with the structural-validity checklist.

### 9.6 Reproducibility and Release

To make "open release" concrete, Table 7 states exactly what is and is not published. The boundary is firm: the ledger, hash chain, timestamp anchor, e-value module, schemas, prompts, and synthetic fixtures are released so the ledger mechanics are reproducible and the H2 contrast protocol is auditable at the prompt-and-aggregate level (the literal arm prompts plus the reported aggregate counts — the empirical cards and per-card decisions are private, per Table 7); the *live scoring engine* (the deterministic gate, confidence scoring, and domain-templating code that runs the funnel — including the config-only domain swap of § Portability (c)) stays in the private research book, so that demonstration is reported rather than reader-runnable; and the *live book* (positions, broker data, paid feeds) is never released, because publishing it would read as "copy these trades," which this project explicitly is not. The repository is `https://github.com/yixingz3/qpop`; its `repro/` directory verifies the released *mechanics* with one command each (tests, ledger verification, tamper demo, timestamp anchor, schema validation, and the paper build). The empirical results in this paper are *documented* as released prompts and sanitized aggregates: a reader can inspect the disclosed protocol and recompute the reported aggregate arithmetic, but cannot audit or rescore individual empirical decisions and cannot independently reproduce the results — the per-card records, arm decisions, audit manifest and labels, funnel ledger, and the private engine that produced them are not public.

| **Artifact** | **Released?** | **Contains** |
|---|---|---|
| Code (package) | Yes | ledger, hash chain, OpenTimestamps anchor, e-value module (experimental), CLI, schema validation |
| Gate + scoring engine | No | deterministic gate, confidence scoring, domain templating (runs the live funnel) |
| Prompts | Yes | sanitized reference templates for source / triage / evaluate / adjudicate / audit; the five H2 arm prompts released verbatim (three recovered pilot originals, two later pre-registered extension arms) |
| Candidate cards | Synthetic sample only | schema-conformant sample cards; the empirical 38-card batch and per-card arm outputs are private |
| Ledger | Synthetic | QPOP schema and sample hashes |
| Live positions | No | not released |
| Broker data | No | not released |
| Paid data | No | not redistributed |

Table 7: What the open release contains.

## 10 Discussion

**Cross-domain portability.** Adding a domain is mostly configuration-driven, and this is now *demonstrated*, not merely intended. The second- and third-domain runs (§Portability) support the *method-level* claim — the same engine located the canonical chokepoint and rejected the conglomerate look-alikes on a different node list, the rejection taxonomy transferred, and a domain-specific bypass mode (customer in-sourcing) was absorbed into the bear case unmodified. The stronger *config-only* capability is demonstrated as well: after threading a `domain` parameter through the SOURCE/EVALUATE templates (byte-identical golden tests pin the live domain), a full uranium-fuel-cycle funnel ran from **one new configuration file and zero prompt-construction code edits** (§Portability, clause (c)) — though the registered H3 conjunction remains unmet, because no single domain satisfies all three clauses. The disciplines (gates, source tiers, bear-case-first, overlap, the forward lock) are domain-agnostic, and the same engine now runs a distinct domain — uranium, one of the rare-earth refining / power-transformer / copper family named above — from a new map alone. The remaining limit is one of *release*, not capability: the templating and scoring engine is private, so the config-only swap is reported here rather than reproducible from the public repository.

**Uncertainty about the constraint (a conceptual case study).** A binding chokepoint is *necessary but not sufficient* for a tradeable edge — the edge lives in the *uncertainty* about the constraint. Physical chokepoints keep pricing power because their magnitude and timing are discovered late; a *synthetic, scheduled* constraint with a known date and a shrinking magnitude is arbitraged toward zero edge. The cleanest illustration is Bitcoin's halving: a real algorithmic supply cut, but the most pre-announced supply event in finance, front-run and with a geometrically decaying marginal shock. This is the same principle as policy-as-evidence — information already in the price is, to that extent, not alpha — and it is why the framework prefers chokepoints whose binding is still being discovered. We present it as an illustration of why *forward, unanticipated* evidence matters, not as a contribution.

**Forced-cheap entry (cross-substrate; see appendix).** The constraint-uncertainty principle above has a testable, cross-substrate form, which we discuss separately in Appendix A: an exploratory, hand-selected historical case series on proprietary-data businesses is *consistent with* the unpriced gap arising from *involuntary forced selling* rather than the market's belated *recognition* of a hidden data business — a hypothesis-generating note, not a backtest, an empirical result, or a performance claim.

**On the illustrations in this section.** The constraint-uncertainty, Bitcoin-halving, and forced-seller discussions above sharpen the framework's concepts (e.g. why the valuation and crowding adjustments exist); they are *not* additional empirical contributions, and should not be read as investment analysis or advice.

## 11 Limitations

The forward sample is small and early; intervals are wide and no live-money track record is claimed. LLM outputs are nondeterministic across model versions, so we log the exact model identifier per run, pin the rendered prompt artifacts by SHA-256 hash in the private provenance record, and treat a version change as a dated amendment. Survivorship enters through which domains get built. Most importantly, the admission rate is *necessary but not sufficient* for forward predictive performance: a disciplined rejection process is a precondition for, not a guarantee of it, and only the pre-registered forward log can establish that. Finally, a screening engine's *coverage* (which chokepoints the SOURCE stage surfaces) is a separate failure mode from its *judgment* (whether the gates correctly accept or reject a surfaced candidate). In the third-domain probe a blind comparison against an external expert reading list revealed a genuine chokepoint the SOURCE stage had missed even though the gates rejected it correctly once it was surfaced — a coverage gap, not a judgment gap. We treat that distinction as load-bearing: a coverage gap is closed by adding a source lens, never by loosening the gates, and we use blind held-out comparisons against external lists as the way to detect coverage gaps when entering a new domain.

**What the ledger proves — and what it does not.** The hash chain establishes *tamper-evidence*: any post-hoc edit, insertion, deletion, or reorder of a closed entry is detectable. It does *not*, on its own, establish *wall-clock time* — a ledger written later could carry a backdated `created` field. A full "registered *before* the outcome" guarantee therefore requires an external timestamp anchor: a signed Git commit or release, a public CI artifact, or a public append-only timestamp service (e.g. OpenTimestamps, or a Sigstore/Rekor transparency-log entry). The released package now *implements* this as an opt-in step — an OpenTimestamps backend with `anchor` / `verify-external` subcommands and a sidecar receipt, covered by 11 hermetic tests plus a dispatch-gated live CI job (Rekor was evaluated and declined on dependency weight). It is **manual and per-invocation**, not automatic on every ledger append, so we still do not claim an automatic cryptographic proof-of-time for each entry; the anchor is available and tested, and applying it remains a deployment choice.

## 12 Ethics and Responsible Use

This is a research framework, not investment advice, and we make no performance claim. The system is optimized for auditability: each admitted position is explainable from its content-hashed ledger entry (we release the schema and synthetic/sanitized examples, not the live ledger). We comply with data-provider terms (no paid feeds are redistributed), publish only synthetic and sanitized examples, and never release live positions, broker credentials, or the scored live map. See `ETHICS.md` and `DISCLAIMER.md` in the repository.

**Conflicts of interest and disclosure.** The author maintains a private research book that applies this method and may hold positions in names within the example domains discussed here. Example names are illustrations of the decision taxonomy — not endorsements, recommendations, or a disclosure of the book; no live positions, weights, or sizes are revealed. Where example cases are named (Appendix A), they are historical, closed dislocations discussed qualitatively as an exploratory case note; current or live candidates are described only generically. The author declares no other competing interests.

**AI disclosure.** See the AI-disclosure statement following the abstract.

## 13 Conclusion

LLMs make thematic research cheap but unreliable as decision engines. We presented a forward-registered, auditable workflow that converts an LLM from a decision engine into a sourcing tool: deterministic gates, source-tiered dated evidence, a bear case written before any recommendation, a hash-chained pre-registration ledger, and forward (not backtest) validation. The headline is restraint — the agent rejects most of what it surfaces. A held-out bull-only LLM auditor agreed with 31/40 (0.775) of a documented, seeded sample of those rejections — *missing* the $\geq 0.80$ criterion later registered for the prospective window — and a full-record adjudicator overrode all nine disagreements (9/9); we report the miss as the honest pilot reading and the override pattern as model disagreement under asymmetric information (an LLM-on-LLM diagnostic, not a human audit — rejection quality remains unresolved pending the pre-specified human lane), and exploratory system contrasts show large admission-rate increases in every weakened-discipline configuration — with no single component's causal contribution identified. The framework scaffold, a worked example, and a domain template are released for others to fork and extend.

# A Cross-Substrate Portability Probe: Proprietary-Data Monopolies

The domains in § Portability vary the node list but share a *physical* substrate — factories, fill-finish lines, launch hardware. As a further portability probe, we applied the same disciplines to a *non-physical* substrate: proprietary data and network monopolies, where the chokepoint is a dataset an industry cannot rebuild (registration and title histories, citation graphs, contributor-fed credit files) rather than a plant. **This appendix is an exploratory, hypothesis-generating case note, not an empirical study.** The probe is design-stage, not live: after drafting the substrate's gates, we examined a small, hand-selected set of well-known historical dislocations in such businesses. No selection protocol, enumerated case universe, point-in-time sourcing standard, valuation definition, or outcome/re-rating protocol was documented, so no number from this exercise is reported as evidence and no generalization is claimed from it. The portability observation the appendix records is qualitative: the same purity and over-payment gates separated genuine data monopolies from contestable look-alikes, and rejected celebrated data businesses that were never actually cheap.

**Forced-cheap entry: the hypothesis this case note generates.** The constraint-uncertainty principle has a testable, cross-substrate form. A chokepoint is mispriced only when its price is *involuntarily* dislocated, and the dislocation mechanism is substrate-specific. Physical chokepoints get cheap on a capital-spending *cycle*; proprietary-data chokepoints compound steadily and rarely cycle cheap, so their plausible forced-cheap window is an *observable forced seller* — a spin-off orphaned by index funds, an index deletion, a private-equity exit. In the hand-selected historical cases we examined, the forced-seller dislocations (a spin-off orphaned by index funds, a post-breach forced sale, a private-equity carve-out) appeared to leave material valuation gaps to information-services peers, while the *recognition* dislocations — the market belatedly discovering a hidden data business — appeared to trade at *premiums*, leaving no gap to capture; recognition is an *anticipated* re-pricing, arbitraged just as the pre-announced halving is. Because the cases were selected with hindsight from celebrated episodes, with no enumerated universe, documented valuation definition, or outcome protocol, these observations are not evidence and we draw no conclusion from them. What they generate is the falsifiable entry-rule hypothesis — *in this substrate, only involuntary dislocations leave an unpriced gap* — which a properly documented, dated, pre-registered protocol would have to test before the framework could rely on it.

# References


45ck. llm-quant: LLM-powered paper trading system with a tamper-evident SHA-256 trade ledger. `https://github.com/45ck/llm-quant`, 2025. open-source GitHub repository; no associated paper.

Mostapha Benhenda. Look-ahead-bench: a standardized benchmark of look-ahead bias in point-in-time LLMs for finance, 2026. URL `https://arxiv.org/abs/2601.13770`.

Abel Brodeur, Nikolai M. Cook, Jonathan S. Hartley, and Anthony Heyes. Do preregistration and preanalysis plans reduce p-hacking and publication bias? evidence from 15,992 test statistics and suggestions for improvement. *Journal of Political Economy Microeconomics*, 2(3):527–561, 2024. doi: 10.1086/730455. URL `https://www.journals.uchicago.edu/doi/10.1086/730455`.

Alessio Buscemi, Tom Deckenbrunnen, Fahria Kabir, Kateryna Mishchenko, and Nishat Mowla. Assessing high-risk AI systems under the EU AI Act: From legal requirements to technical verification, 2026. URL `https://arxiv.org/abs/2512.13907`.

Katherine Casey, Rachel Glennerster, and Edward Miguel. Reshaping institutions: Evidence on aid impacts using a preanalysis plan. *Quarterly Journal of Economics*, 127(4):1755–1812, 2012. doi: 10.1093/qje/qje027. URL https://academic.oup.com/qje/article-abstract/127/4/1755/1841616.

Yanxu Chen, Zijun Yao, Yantao Liu, Amy Xin, Jin Ye, Jianing Yu, Lei Hou, and Juanzi Li. StockBench: Can LLM agents trade stocks profitably in real-world markets?, 2025. URL https://arxiv.org/abs/2510.02209.

Han Ding, Yinheng Li, Junhao Wang, Hang Chen, Doudou Guo, and Yunbai Zhang. Large language model agent in financial trading: A survey, 2024. URL https://arxiv.org/abs/2408.06361.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL https://arxiv.org/abs/2305.14325.

Sebastian Fritz-Morgenthal, Bernhard Hein, and Jochen Papenbrock. Financial risk management and explainable, trustworthy, responsible AI. *Frontiers in Artificial Intelligence*, 5: 779799, 2022. doi: 10.3389/frai.2022.779799. URL https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2022.779799/full.

Zhenyu Gao, Wenxi Jiang, and Yutong Yan. Detecting lookahead bias in LLM forecasts, 2025. URL https://arxiv.org/abs/2512.23847.

Campbell R. Harvey, Yan Liu, and Heqing Zhu. ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68, 2016. doi: 10.1093/rfs/hhv059. URL https://academic.oup.com/rfs/article/29/1/5/1843824.

Jake M. Hofman, Angelos Chatzimparmpas, Amit Sharma, Duncan J. Watts, and Jessica Hullman. Pre-registration for predictive modeling, 2023. URL https://arxiv.org/abs/2311.18807.

Kexin Huang, Ying Jin, Ryan Li, Michael Y. Li, Emmanuel Candès, and Jure Leskovec. Automated hypothesis validation with agentic sequential falsifications, 2025. URL https://arxiv.org/abs/2502.09858. ICML 2025.

Raffi Khatchadourian. Replayable financial agents: A determinism-faithfulness assurance harness for tool-using LLM agents, 2026. URL https://arxiv.org/abs/2601.15322. ICLR 2026 Workshop on Advances in Financial AI.

Yaxuan Kong, Hoyoung Lee, Yoontae Hwang, Alejandro Lopez-Lira, Bradford Levy, Dhagash Mehta, Qingsong Wen, Chanyeol Choi, Yongjae Lee, and Stefan Zohren. Evaluating LLMs in finance requires explicit bias consideration, 2026. URL https://arxiv.org/abs/2602.14233.

Hoyoung Lee, Junhyuk Seo, Suhwan Park, Junhyeong Lee, Wonbin Ahn, Chanyeol Choi, Alejandro Lopez-Lira, and Yongjae Lee. Your AI, not your view: The bias of LLMs in investment analysis, 2025. URL https://arxiv.org/abs/2507.20957. ACM ICAIF 2025.

Dawei Li, Bohan Jiang, Liangjie Huang, Alimohammad Beigi, Chengshuai Zhao, Zhen Tan, Amrita Bhattacharjee, Yuxuan Jiang, Canyu Chen, Tianhao Wu, Kai Shu, Lu Cheng, and Huan Liu. From generation to judgment: Opportunities and challenges of LLM-as-a-judge. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025a. URL https://arxiv.org/abs/2411.16594.

Xiangyu Li, Yawen Zeng, Xiaofen Xing, Jin Xu, and Xiangmin Xu. Profit mirage: Revisiting information leakage in LLM-based financial agents, 2025b. URL https://arxiv.org/abs/2510.07920.

Dong Lou and Christopher Polk. Comomentum: Inferring arbitrage activity from return correlations. *Review of Financial Studies*, 35(7):3272–3302, 2022. URL https://academic.oup.com/rfs/article-abstract/35/7/3272/6412574.

Chiyu Ma, Enpei Zhang, Yilun Zhao, Wenjun Liu, Yaning Jia, Peijun Qing, Lin Shi, Arman Cohan, Yujun Yan, and Soroush Vosoughi. Judging with many minds: Do more perspectives mean less prejudice? on bias amplifications and resistance in multi-agent based LLM-as-judge, 2025. URL https://arxiv.org/abs/2505.19477.

Lars Malmqvist. Sycophancy in large language models: Causes and mitigations, 2024. URL https://arxiv.org/abs/2411.15287.

R. David McLean and Jeffrey Pontiff. Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32, 2016. doi: 10.1111/jofi.12365. URL https://onlinelibrary.wiley.com/doi/abs/10.1111/jofi.12365.

Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114. URL https://www.pnas.org/doi/10.1073/pnas.1708274114.

Lingfei Qian, Xueqing Peng, Yan Wang, Vincent Jim Zhang, Huan He, Hanley Smith, Yi Han, Yueru He, Haohang Li, Yupeng Cao, Yangyang Yu, Alejandro Lopez-Lira, Peng Lu, Jian-Yun Nie, Guojun Xiong, Jimin Huang, and Sophia Ananiadou. When agents trade: Live multi-market trading benchmark for LLM agents, 2025. URL https://arxiv.org/abs/2510.11695. introduces the Agent Market Arena (AMA) benchmark.

Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023. doi: 10.1214/23-STS894.

Anisha Roy and Dip Roy. MemGuard-Alpha: Detecting and filtering memorization-contaminated signals in LLM-based financial forecasting via membership inference and cross-model disagreement, 2026. URL https://arxiv.org/abs/2603.26797.

Glenn Shafer. Testing by betting: A strategy for statistical and scientific communication. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 184(2):407–431, 2021. doi: 10.1111/rssa.12647.

Valerio Volpati, Michael Benzaquen, Zoltan Eisler, Iacopo Mastromatteo, Bence Toth, and Jean-Philippe Bouchaud. Zooming in on equity factor crowding. *Journal of Risk*, 23(1), 2020. URL https://arxiv.org/abs/2001.04185.

Vladimir Vovk and Ruodu Wang. E-values: Calibration, combination and applications. *The Annals of Statistics*, 49(3):1736–1754, 2021. doi: 10.1214/20-AOS2020.

Shijie Wu, Ozan Irsoy, Steven Lu, Vadim Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhanjan Kambadur, David Rosenberg, and Gideon Mann. BloombergGPT: A large language model for finance, 2023. URL https://arxiv.org/abs/2303.17564.

Yijia Xiao, Edward Sun, Di Luo, and Wei Wang. TradingAgents: Multi-agents LLM financial trading framework, 2024. URL https://arxiv.org/abs/2412.20138.

Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. FinGPT: Open-source financial large language models, 2023. URL https://arxiv.org/abs/2306.06031.

Yuxuan Ye, Jun Han, Ao Hu, Juncheng Bu, Yiyi Chen, Liangjian Wen, Danilo Mandic, Danny Dongning Sun, Xu Yinghui, and Zenglin Xu. The alpha illusion: Reported alpha from LLM trading agents should not be treated as deployment evidence, 2026. URL https://arxiv.org/abs/2605.16895.

Wentao Zhang, Lingxuan Zhao, Haochong Xia, Shuo Sun, Jiaze Sun, Molei Qin, Xinyi Li, Yuqing Zhao, Yilei Zhao, Xinyu Cai, Longtao Zheng, Xinrun Wang, and Bo An. A multimodal foundation agent for financial trading: Tool-augmented, diversified, and generalist, 2024. URL https://arxiv.org/abs/2402.18485.

Zhenyu Zhao, Aparna Balagopalan, Adi Agrawal, Dilshoda Yergasheva, Waseem Alshikh, and Daniel M. Bikel. The price of agreement: Measuring LLM sycophancy in agentic financial applications, 2026. URL https://arxiv.org/abs/2604.24668. ICLR 2026 Workshop on Financial AI (FinAI).